

a variety of popular programming languages. When MySQL was written, it was composed in C and C++, and is wholly optimized for both the Unix and Win32 platforms. MySQL also uses memory hash tables, kernel threads, certain high optimized individual collected class libraries and thread based memory portions.

MySQL supports different field types including CHAR, FLOAT, DOUBLE, DATE, VARCHAR, TEXT, SET, BLOB and ENUM. Certain advanced querying and grouping functions such as, COUNT(), GROUP BY and ORDER BY, STD(),AVG(),MAX(),MIN() and SUM() are also supported here.

11.1 Database concept

The term 'data' suggests a record with certain necessary information. A database in computer is a structured record collection. They are stored in computer systems. A database structure is prepared by arranging a data in a database model patterns. There are three types of databases that are frequently used:

- Relational model
- Hierarchical model
- Network model

A database is organised in a computer with a database management system. Using this software, a computer performs different related functions. These include recovering data, storing, adding, deleting and modifying data.

An extensive part of a database depends on different managing factors like integrity, presentation, concurrency and recovery from hardware failures. We can divide a database management system into two categories - desktop databases and server databases. The desktop databases are generally targeted towards individual user application. The server databases target towards the authenticity and uniformity of a data.

A database is basically of two types - flat file and relational database.

Flat file: Flat file databases usually store small amounts of data. They are easily read and edited. Basically, they are arranged in a series and are accordingly analysed. Flat files are best to store sim-